National University of Computer and Emerging Sciences
Lahore Campus

Signals & Systems (EE2008)

Final Exam

Date: May 22, 2024

Course Instructor(s)

Mr. Dr. S.M. Sajid

Mr. Asim Hussain Farooqi

Total Time (Hrs):	3
Total Marks:	100
Total Questions:	6

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Roll No   Section   Student Signature

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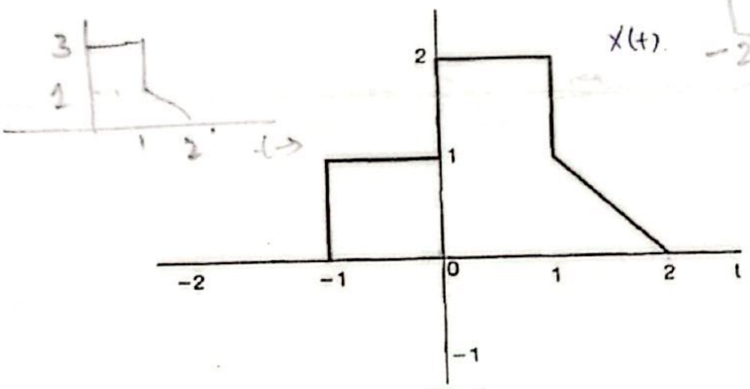
Attempt all the questions.

CLO # 1: Express different types of signals and systems.

Q1: A continuous-time signal x(t) is shown in Fig-1 below. **Compute** and label carefully each of the following signals: **[20]**

a) $x(2t+1)$

b) $[x(t)+x(-t)]u(t)$



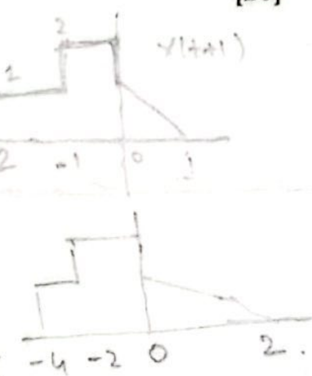
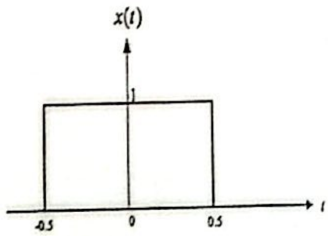
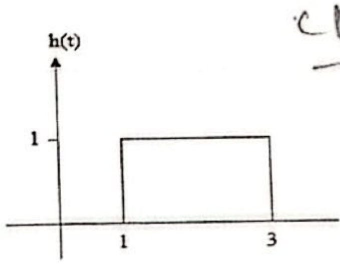


Fig-1

CLO # 2: Apply mathematical and graphical convolution techniques to analyze CT and DT systems.

Q2: Solve the convolution by graphical method, the signal x(t) and h(t) are given below. **[20]**





$$y(t) = \begin{cases} t-0.5 & 0.5 \le t \le 1.5 \\ 1 & 1.5 \le t \le 2.5 \\ -t+3.5 & 2.5 \le t \le 3.5 \end{cases}$$

CLO # 3: Analyze signals using Fourier series representation.

Q3: Analyze the periodic signal shown in Fig-2, calculate all the coefficients of trigonometric Fourier series and also write the final expression of trigonometric Fourier series. **[20]**

$a_0 = 0.5$

$a_n = 0.$

$b_n = \frac{-1}{n\pi}$

$T_0 = 2\pi$

$\omega_0 = 1$

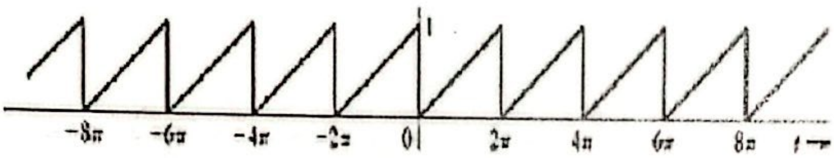


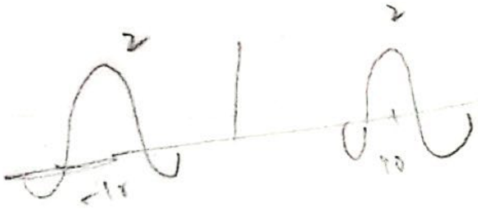
Fig-2

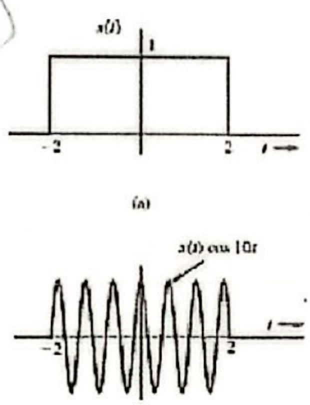
$x(t) = \frac{1}{2} + \frac{1}{\pi}\sin t - \frac{1}{2\pi}\sin 2t - \frac{1}{3\pi}\sin 3t - \dots$

CLO # 4: Analyze signals using Fourier transform and its properties.

Q4: Analyze the signal given below, find and sketch the Fourier transform of the modulated signal $x(t)\cos(10t)$ in which $x(t)$ is a gate pulse $rect\left(\frac{t}{4}\right)$ as illustrated in figure given below. **[15]**

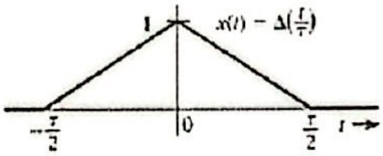
$2\{\text{sinc}(2(\omega-10))+\text{sinc}(2(\omega+10))\}$





CLO # 4: Analyze signals using Fourier transform and its properties.

Q5: Analyze the signal given below, find and sketch the Fourier transform of $\Delta\left(\frac{t}{T}\right)$ clearly mention the method used or the property applied. **[15]**



$\frac{T}{2}\,\text{sinc}^2\left(\frac{\omega T}{4}\right)$

CLO # 5: Describe and use the sampling theorem.

Q6: State and prove the sampling theorem. **[10]**